

Research and Ethics in Agricultural and Resource Economics

First Years

Part 1

Anna Josephson & Jeffrey Michler

Adapted from April Athnos (MSU), Travis Lybbert (UC Davis), and Sarah Janzen (UIUC)

What is research?

- Research is a process to discover new knowledge.
- National Academy of Sciences: the objective of research is to “extend human knowledge of the physical, biological, or social world beyond what is already known.”



Social science v. “Bench science”

- “Bench science”: physics, biology, chemistry
 - Lab-based
- Social science: economics
 - Surveys
 - Experiments
 - Quantitative and qualitative data

soft science v. hard science

Ag. and Resource Economics

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ethic*

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Journal Article

The Ethics and Economics of Agrifood Competition

Michael Gunderson

American Journal of Agricultural Economics, Volume 96, Issue 2, 1 March 2014, Pages 611–612,
<https://doi.org/10.1093/ajae/aat112>

Published: 16 January 2014

...Michael Gunderson mgunders@purdue.edu James, Harvey. The **Ethics** and Economics of Agrifood Competition . Springer, Dordrecht, 2013. 240 pp. \$129.00. © The Author (2014). Published by Oxford University Press on behalf of the Agricultural and Applied Economics Association. All rights reserved...

[Extract](#) ▼

Journal Article

Risk Perceptions And Preferences For Ethical And Safety Credence Attributes

Ellen W. Goddard, Jill E. Hobbs, Brian G. Innes, Patrycja E. Romanowska, Adrian D. Uzea

American Journal of Agricultural Economics, Volume 95, Issue 2, 1 January 2013, Pages 390–396,
<https://doi.org/10.1093/ajae/aas108>

Published: 01 January 2013

... grouped under the broad heading of **ethical** preferences, and analysts often relate these attributes strongly to public good aspects of food production (and altruistic preferences) (Lusk, Nilsson, and Foster 2007 ; Umberger, Thilmany, McFadden, and Smith 2009). Are these so-called **ethical** preferences...

[Extract](#) ▼

Ethics is often not discussed in agricultural & resource economics.

But, it is essential to:

- ...train new researchers in ethics
- ...discuss the details and subtleties of misconduct
- ...address how economists make day-to-day ethical decisions
- ...ensure that our work is replicable and reproduceable

Replication Crisis

A 2016 Nature poll of 1,500 scientists reported that 70% of respondents had failed to reproduce at least one other scientist's experiment.

- That included: 87% of chemists, 77% of biologists, 69% of physicists and engineers, 67% of medical researchers, 64% of earth and environmental scientists, and 62% of all others.
- 50% had failed to reproduce one of their own experiments.
- Less than 20% had ever been contacted by another researcher unable to reproduce their work.
- A minority had ever attempted to publish a replication, and of those 24% had been able to publish a successful replication, while 13% had published a failed replication.

In 2009, 2% of scientists admitted to falsifying studies at least once and 14% admitted to personally knowing someone who did.

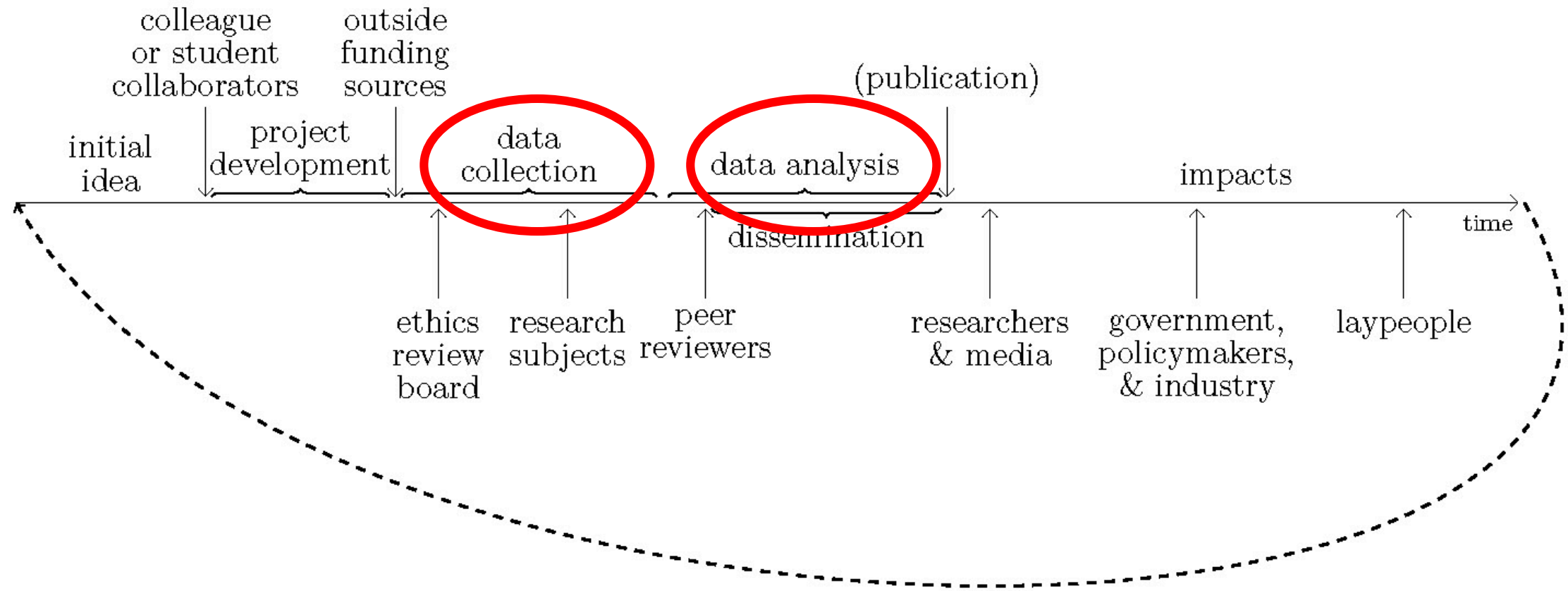
Replication Crisis in Economics

A 2016 study in the journal *Science* found that one-third of 18 experimental studies from two top-tier economics journals (*American Economic Review* and the *Quarterly Journal of Economics*) failed to successfully replicate.

- With the original researchers' assistance, that percentage increased to about half, suggesting reporting practices and requirements are seriously deficient.

A 2017 study in the *Economic Journal* suggested that "the majority of the average effects in the empirical economics literature are exaggerated by a factor of at least 2 and at least one-third are exaggerated by a factor of 4 or more".

Life Cycle of Research Project



Unethical Data Collection

- Tuskegee Syphilis Experiment
- Henrietta Lacks
- Sanford Prison Experiment

And the “solution”:

- Institutional Review Boards



Tuskegee Syphilis Experiment

- Conducted between 1932 and 1972 by the U.S. Public Health Service
- Purpose was to study the natural history of untreated syphilis
- African-American men in the study were told they were receiving free health care
- *But their syphilis was left untreated*



Henrietta Lacks



- August 1, 1920 – October 4, 1951
- African-American woman treated at Johns Hopkins Hospital for cervical cancer
- Hospital cultured cells collected during a biopsy of a tumor
- Researchers immortalized these cells, now called HeLa and used for medical research and for commercial purposes
- Lacks **never provided her consent**, her family receives no money from the commercial use of these cells

Stanford Prison Experiment

- Social psychology experiment conducted with college students from August 14–20, 1971
- Attempted to investigate the psychological effects of perceived power
- Students randomly assigned to “guards” or “prisoners” in a mock prison
- Participants played their “roles” too well
- “Guards” ended up physically and psychologically abusing “prisoners”



Institutional Review Boards

- **IRB:** administrative body created to protect the rights and welfare of human research participants.
 - Belmont Report (1979) set requirements for research with human subjects
 - Define processes and procedures required to conduct research.
- Josephson and Michler (2018):
 - 1. Discrepancy between requirements and practice
 - 2. Use primarily within ‘western’ universities

Human Subjects Protection Program

Welcome to the University of Arizona Human Subjects Protection Program

The Human Subjects Protection Program (HSPP), as the administrative and regulatory support program to the Institutional Review Boards (IRBs), works in collaboration with the research community to maintain an ethical and compliant research program. The IRBs are the independent review committee charged with the protection of human research subjects. An IRB must review all research and related activities involving human subjects conducted at the University of Arizona or by in which the University is a responsible participant. The University of Arizona HSPP has been accredited by the Association for Accreditation of Human Research Protection Programs (AAHRPP) since 2005, strong evidence of our commitment to the protection of human research subjects.



REPORT MISCONDUCT

University Ethics and Compliance Hotline:

866-364-1908

HSPP CONTACTS

Department email:

VPR-IRB@email.arizona.edu

Subscribe to the **HSPP listserv*

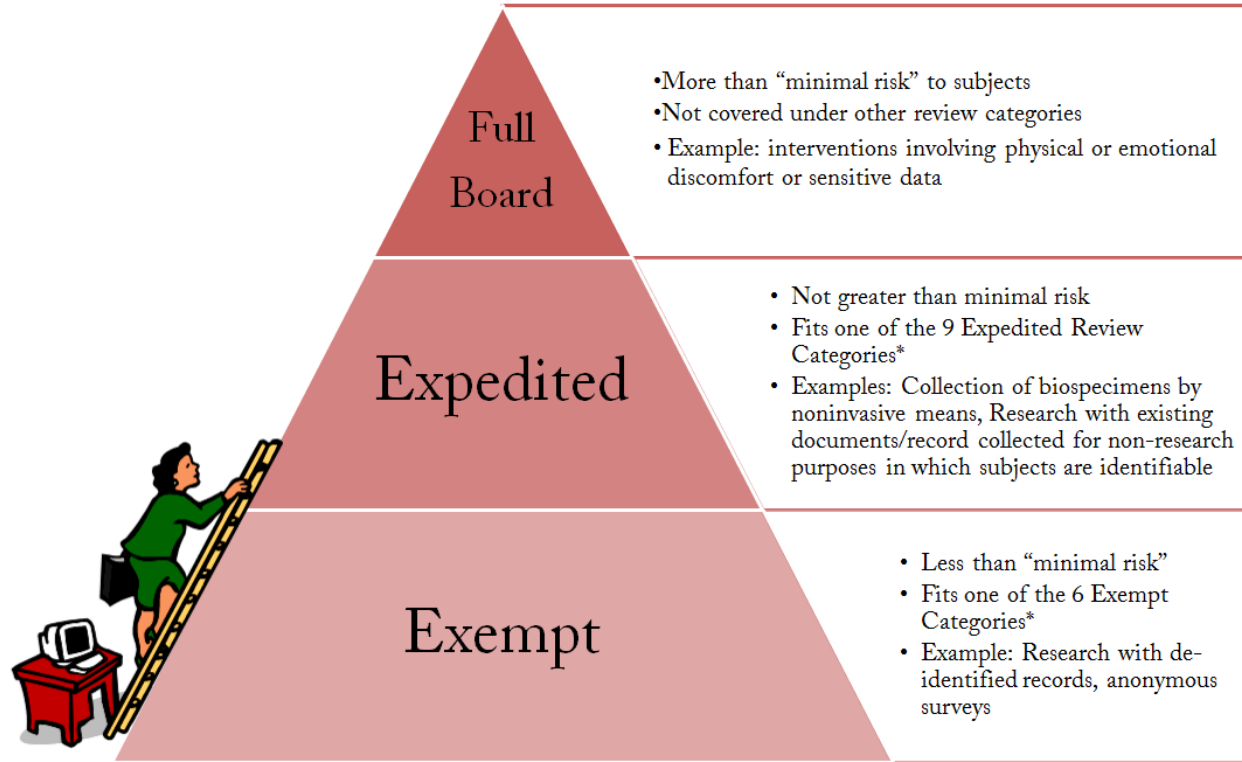
Mariette Marsh, Director
(520) 626-7575

Christine Melton-Lopez, Assistant
Director
(520) 626-8630

Mason Burchfield, IRB Coordinator
(520) 626-0433

Most economic studies are *exempt*

Levels of IRB Review



*Defined by federal regulation (45 CFR 46)

- Research can be approved as “exempt” if it is no more than “**minimal risk**” and fits one of the exempt review categories as defined by federal regulation.

Content to participate



- Requirement that researchers explain any risks associated with participation in a study to those involved
- Informed consent is obtained before research activity takes place
 - Voluntary
 - Informed of risks
 - Privacy maintained

View of IRB ?

- *IRBs exist to protect institutions from legal repercussions*

Unclear if IRBs succeed in **stated goal** of protecting human participants

- Particularly when those subjects are on a different continent

Unethical Data Management and Analysis

- Michael Lacour
- Dan Ariely
- Brian Wansink

And the “solution”???:

- ...to come in the next session !



“If you don't reveal some insights soon, I'm going to be forced to slice, dice, and drill!”

Michael LaCour

- Falsified data: "When Contact Changes Minds: An Experiment on Transmission of Support for Gay Equality."
 - *Science*, Media: NPR, NYTimes...
- Falsified grants: \$793,000
- Falsified analysis



Last month, the community was shaken when a major study on gay marriage in Science was retracted following questions on its funding, data, and methodology. The senior author, Donald Green, made it clear he was not privy to many details of the paper — which raised some questions for C. K. Gunsalus, director of the National Center for Professional and Research Ethics, and Drummond Rennie, a former deputy editor at JAMA. We are pleased to present their guest post, about how co-authors can carry out their responsibilities to each other and the community.

Just about everyone understands that even careful and meticulous people can be taken in by a smart, committed liar. What's harder to understand is when a professional is fooled by lies that would have been prevented or caught by adhering to community norms and honoring one's role and



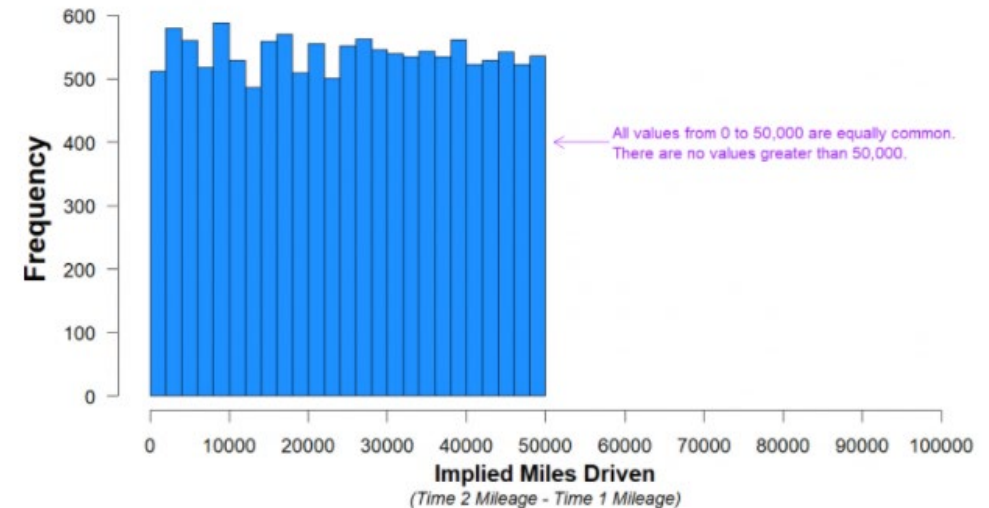
C. K. Gunsalus

Dan Ariely (???)

- Recent, recent, recent: August 2021.
- 13,500 drivers were randomly sent one of two policy review forms to sign:
 - “I promise that the information I am providing is true” appeared at the *bottom* of the document and the other where the statement appear at the *top*.
- Falsified data.
- Falsified analysis?



<https://datacolada.org/98>



Brian Wansink

“The Grad Student Who Never Said ‘No’”

- “I told her to work hard [and] squeeze some blood out of this rock.”
- “She analyzed the data over and over until she began discovering solutions that held up.”



Wansink Unwittingly Triggers Mass Reexamination

- Based on his Nov 2016 blog post, a few start stress testing the published articles
- Discrepancies and suspicious patterns – like “blood in the water”

“Scrutiny of Wansink's work began after a 2016 blog post in which he recounted giving a data set to a graduate student and telling her there's **"got to be something here we can salvage."** That caught the attention of Dutch graduate student Tim van der Zee and others, who subsequently began reviewing Wansink's work and finding errors.”

-NY Times 9-20-2018



- John S. Dyson Chair of Consumer Behavior
- Director: Cornell Food and Brand Lab
- Co-Director of:
 - Institute for Behavioral Economics and Consumer Choice
 - Cornell Center for Behavioral Economics in Child Nutrition Programs

Popularity and Publication $\neq$ Immunity

- Now that the dust has (mostly) settled...
 - 18 of his articles retracted with many more “corrected” (one more than once)
 - Co-author Collin Payne left New Mexico State University in Jan 2018 with email records showing “p-hacking on steroids” E.g., “*Do not despair. It looks like stickers on fruit may work (with a bit more wizardry).*”
 - Cornell issues findings in Sep 2018: “Professor Wansink committed academic misconduct in his research and scholarship, including misreporting of research data, problematic statistical techniques, failure to properly document and preserve research results, and inappropriate authorship.” → Wansink resigns
- The Great Recession was a watershed moment of reflection in macro and finance
- Wansink’s actions and the wider ‘replication crisis’ offers a similar opportunity for professional reflection and introspection in applied microeconomics

Data Manipulation: p -Hacking or HARKing

Manipulate the data or specification so that the **result you want** becomes significant.

1. *p -Hacking / fishing / data dredging / data mining / researcher degrees of freedom* – variable or sub-sample construction, specifications.
 - e.g. “Do not despair. It looks like stickers on fruit may work (with a bit more wizardry).”
2. *Hypothesizing After Results are Known (HARKing)* – running results to see what is significant, without a hypothesis as your guide.
 - Not guided by theory or conceptual model.

Defining and Describing Data Manipulation

- Leamer: “Let’s take the ‘con’ out of econometrics” (1983).
- Kennedy: “Sinning in the basement” (2002).
- Gelman and Loken: “Garden of forking paths” (2013):
“...dataset can be analyzed in so many different ways (with the choices being not just what statistical test to perform but also decisions on what data to [include] or exclude, what measures to study, what interactions to consider, etc.), that very little information is provided by the statement that a study came up with a $p < .05$ result.”

Falsification and Fabrication

Falsification is manipulating research data with the intention of giving a false impression.

- This includes manipulating images, removing outliers or “inconvenient” results, changing, adding or omitting data points, etc.

Similar to fabrication, which includes the wholesale making up of data.

There is a general perception that falsification and fabrication are “worse”.

- But falsification, *p*-Hacking, HARKing, and other methods of data manipulation all serve to undermine the scientific record.

Perspectives on Research Misconduct Motives

Gary Becker's consequentialist view: you should p-hack if:

$$E(\textit{benefit}) > E(\textit{cost}) = \textit{Pr}(\textit{caught}) * \textit{penalty}$$

Diederik Stapel:

- *“Nobody ever checked my work. They trusted me...I did everything myself and next to me was a big jar of cookies...All I had to do was take it.”*

Dan Ariely's “fudge factor” theory of honesty:

- Many (most?) have a strong preference for cheating in small ways, so they can still consider themselves basically honest.
- THIS IS THE GUY FROM A FEW SLIDES AGO!

Perspectives on Research Misconduct Motives

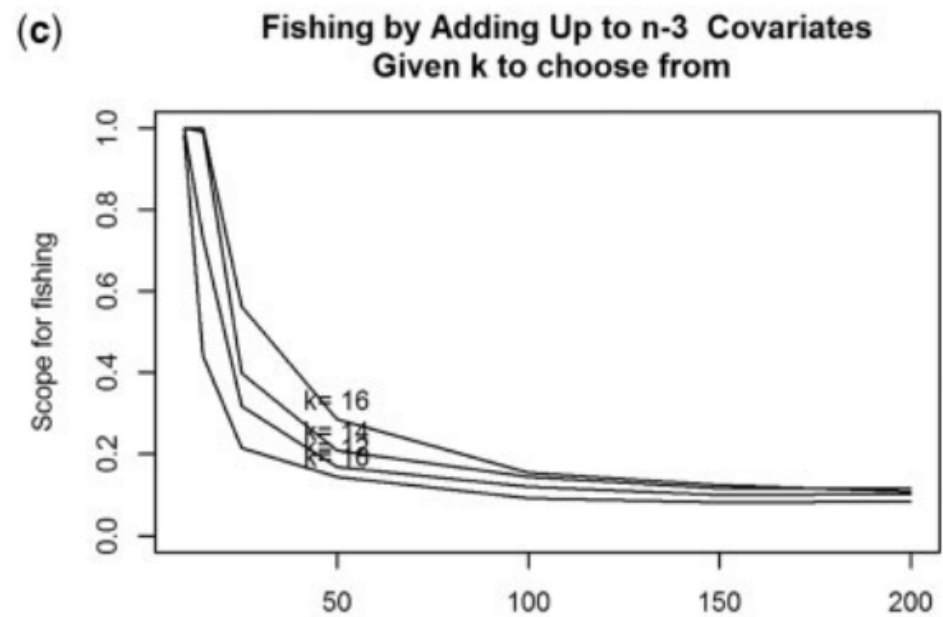
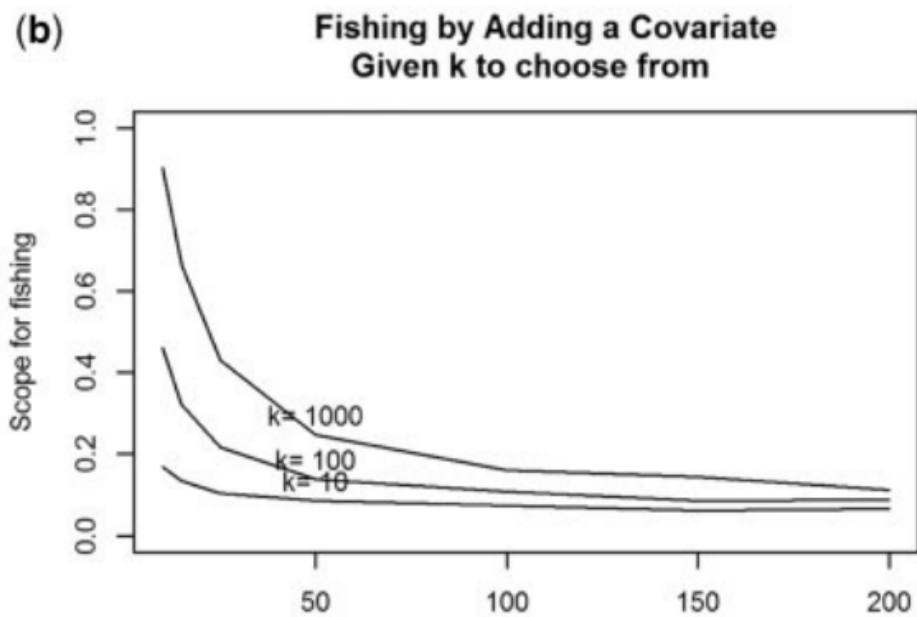
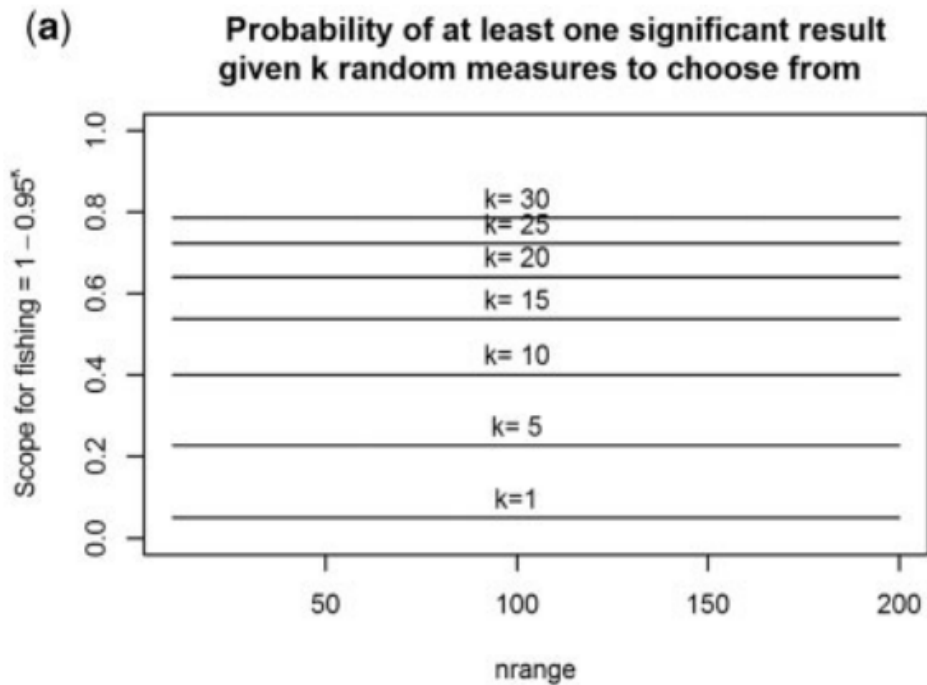
Andrew Gelman – a moralistic framing of these issues is unhelpful and misses the fact that most researchers live in a “statistical fantasy world” in which $p < 0.05$ is the norm:

- *“It’s fine to encourage good behavior and slam bad behavior—but let’s remember that lots of bad work is being done by good people.”*
- *“The problem is a fundamental lack of understanding. Yes, cheating occurs, but the cheating arises out of statistical confusion... The big problem is not overt ‘manipulation’ but researchers fooling themselves.”*

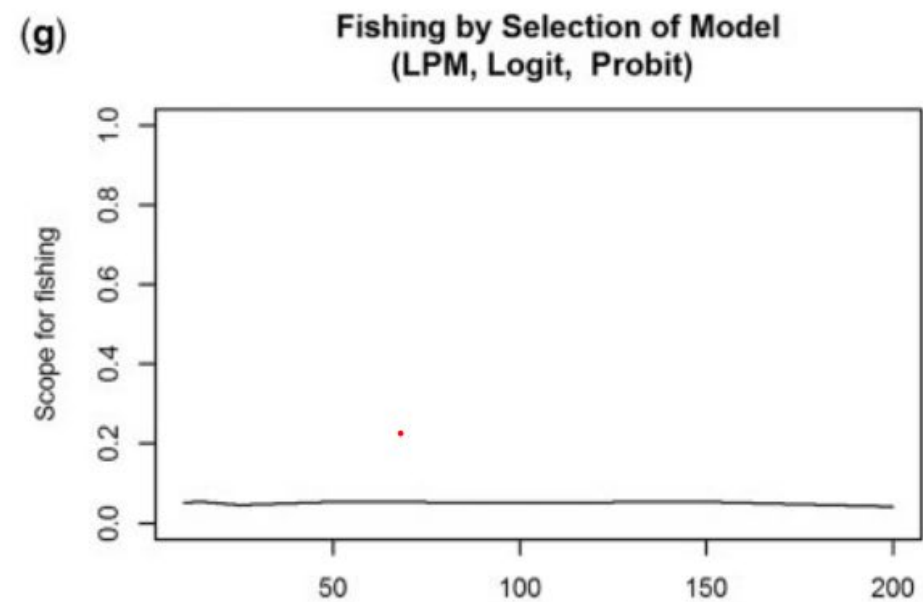
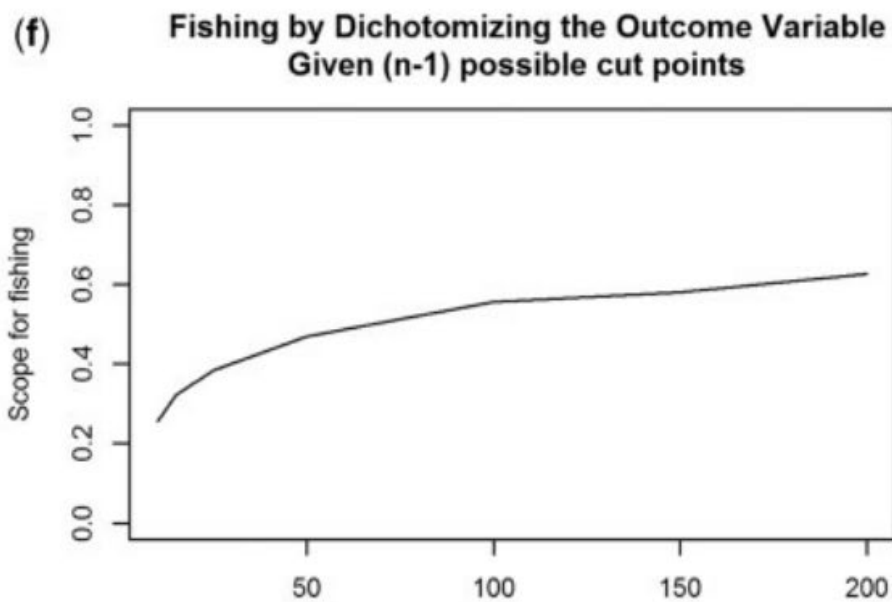
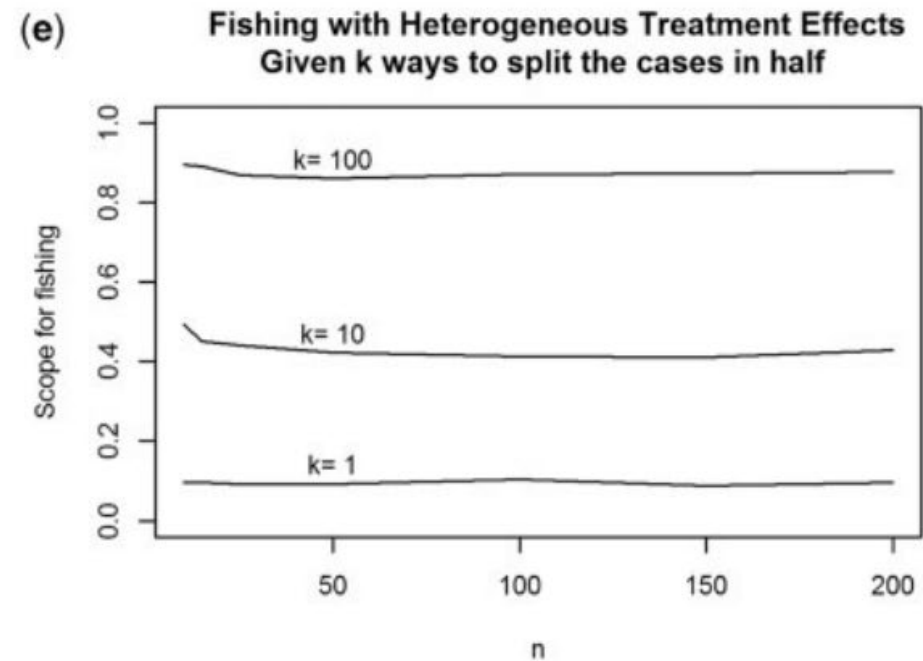
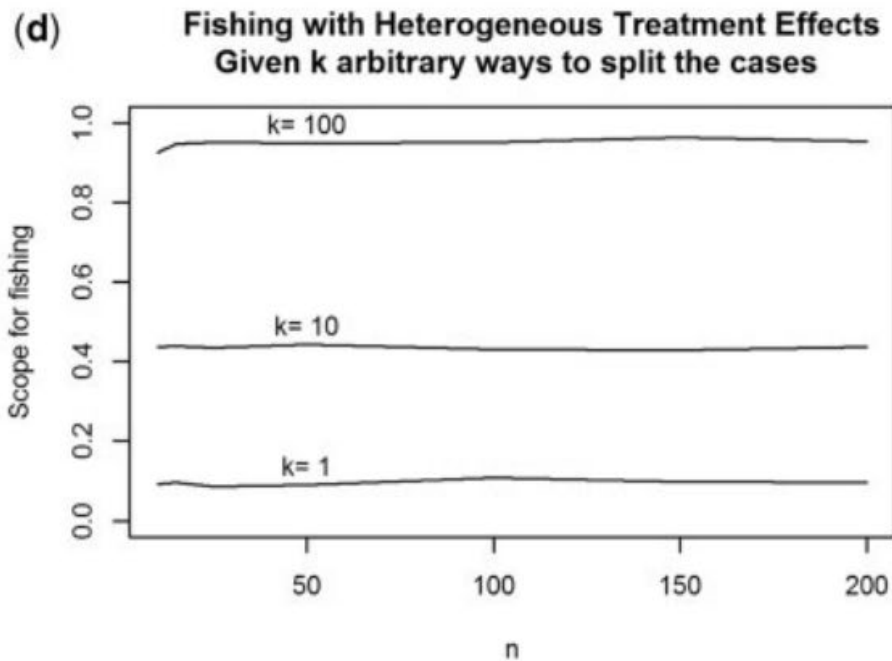
Publication and Manipulation

- Humphreys, Sanchez de la Sierra, van der Windt (2013): using simulation, find that many outcome measures and heterogeneous treatment effects can be used to generate a false positive, even with large sample sizes.
- Brodeur et al. (2016): observe a two-humped density function of test statistics.

Humphreys, Sanchez de la Sierra, van der Windt (2013)



Humphreys, Sanchez de la Sierra, van der Windt (2013)



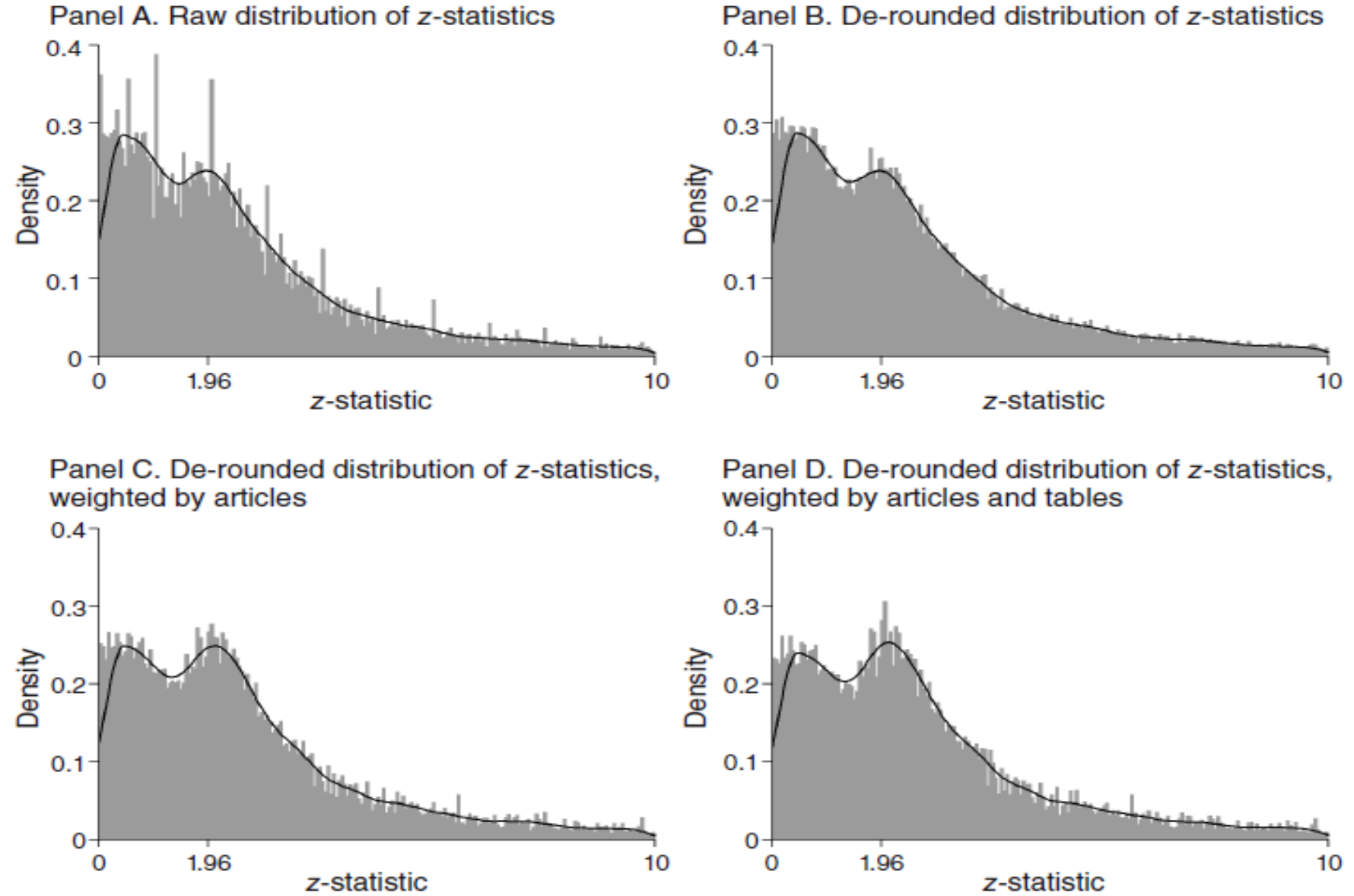
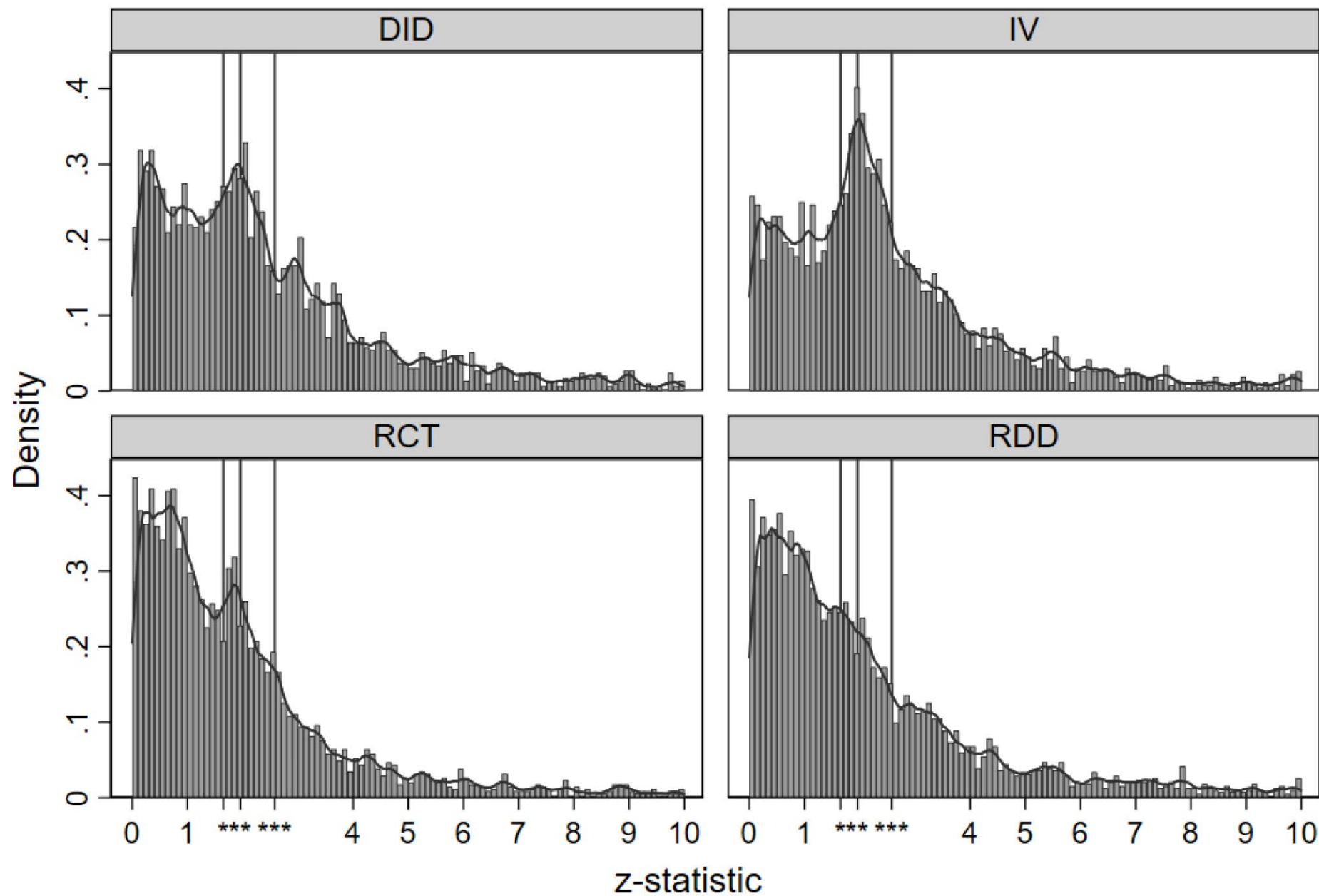


FIGURE 1. DISTRIBUTIONS OF z -STATISTICS

Brodeur et al. (2016)



Graphs by Method

How Can We Change Incentives ?

- 1) Pre-commitments to “tie hands”
- 2) Transparency and openness

More on this in the next session !